

Cool-Colored Roofs

Until recently, cool roofing products came only in light shades. Now a wider range of colors and materials are available.

by Rachel Reiss Buckley

After a hurricane, many homeowners find themselves facing a reroofing job. What these homeowners often don't realize is that the choice of roofing materials will impact a homeowner's energy bills for years, if not decades, to come. Cool roofs—those constructed with reflective materials that limit solar heat gain—cut cooling loads up to 20%.

A conventional cool roof features light-colored surfaces—shades that commercial facilities with flat or low-slope roofs find acceptable. However, homeowners have not adopted the cool-roof approach because they typically prefer the aesthetics of darker colors for their steep-slope roofs. Until recently, there were no cool roofing products available in darker, more appealing hues. The Solution Manufacturers have now developed dark-colored pigments for roofing materials that reflect sunlight instead of absorbing it. These pigments are now being used in coatings for metal roofs, in clay and concrete tiles, and in the multicolored granules that shingles are made of.

Cool-colored roofing products look like their standard counterparts, but they reflect more sunlight and stay cooler (see Figures 1 and 2). Cool-colored coatings for metal roofs are commercially available, and the first cool-colored shingle, more applicable to the residential market, has recently been introduced. In addition, cool coatings for retrofit tiles have been developed by American Roof Tile Coatings; these tiles can be used to replace older tile roofs that have lost their color. Homeowners can also use these tiles if they simply wish to convert to a cool roof. More shingle and tile products are under development.

Solar reflectance—the percentage of the sun's incident radiation that the sur-



MARLIN STUDIOS

This digital image by Tom Marlin of Marlin Studios illustrates cool-colored roofing materials in a variety of styles.

R=0.41	R=0.44	R=0.44	R=0.48	R=0.46	R=0.41
black	blue	gray	terracotta	green	chocolate
R=0.04	R=0.18	R=0.21	R=0.33	R=0.17	R=0.12

BNL

Figure 1. Cool-colored roofing products look like their standard counterparts, but they reflect more sunlight and stay cooler.

face of a material reflects—is an indicator of how cool a roof will be. The higher the solar reflectance of the material, the cooler the roof. With funding from the Public Interest Energy

Research (PIER) program, a few manufacturers have developed dark-colored shingles for use on steep-slope roofs that offer a reflectance of 25% or more. Traditional dark-colored, steep-slope mate-

rials have solar reflectances of approximately 4%–18%, whereas a light-colored cool roof has a reflectance of about 70%. Elk Roofing, using granules developed by 3M, offers the first commercially available cool-colored shingle—a blue-gray product with 25% reflectance. Certain-teed, another manufacturer, has developed a chestnut brown shingle with 40% reflectance, although it is not yet commercially available.

Some cool-colored roofing products may qualify as cool roofs in the California building energy code (the 2005 Title 24 California Building Energy Efficiency standard), which took effect on October 1, 2005. To qualify for compliance credit as a cool roof under the new code, a roofing product must have a minimum initial solar reflectance of 70%. An exception is made for clay tiles to be used on residential buildings, which must have a minimum initial solar reflectance of 40%. Cool roofs will be revisited in the 2008 Title 24 standards, which may include incentives for using colored materials with lower levels of reflectance.

Field Testing

To help manufacturers ensure that cool roofing products will stand the test of time, the PIER-funded cool-colored roofing materials program provides field testing and analysis services. Oak Ridge National Laboratory (ORNL) has established a steep-slope test assembly at its facility in Tennessee to evaluate samples from manufacturing partners. Researchers are measuring the changes in physical composition and appearance that occur as a result of exposure to ultraviolet light, weathering, and temperature changes. A team from Lawrence Berkeley National Laboratory (LBNL) and ORNL has also set up seven test sites in six different climate zones in California. Situated in locations

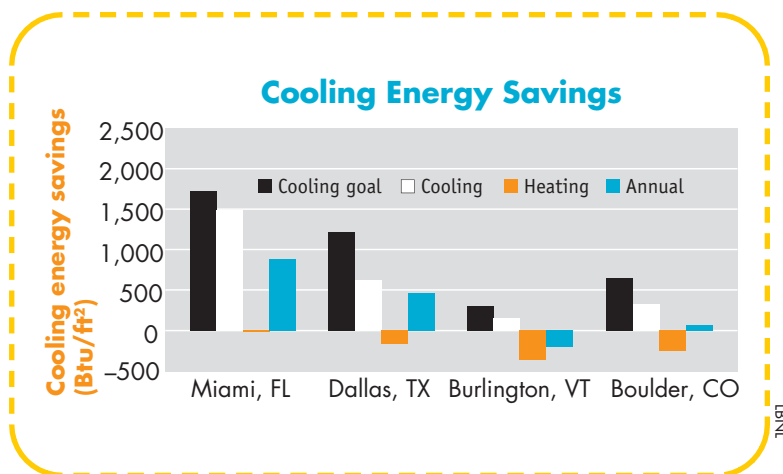


Figure 2. Although cool roofs can increase heating energy in some cold climates, the cooling energy savings lead to overall energy savings in most climates.

ranging from the far north to the extreme south, these sites experience climates ranging from mild to severe. The labs will set up tests at these sites using materials from all of the participating manufacturers. The Sacramento Municipal Utility District is also measuring energy savings and changes in indoor temperature and humidity at test houses equipped with cool roofs in its service territory.

There are two good sources of information about cool-colored roofing materials. LBNL has developed a Web-based database that describes 233 cool-colored pigments and the roofing products that contain each pigment. DOE's Energy Star program maintains a list of cool-roof materials and their manufacturers, but it doesn't verify the accuracy of the information. The Energy Star rating system and material testing is managed by the Cool Roofs Rating Council (CRRC). To qualify for the Energy Star Roof Products list, a steep-slope product must have an initial solar reflectance of greater than or equal to 25% and maintain a reflectance of at least 15% three years after installation.

What's Next?

Manufacturers will continue to develop new pigments for shingle granules and tile coatings. Researchers are also focusing on quantifying the benefits of cool-colored roofing materials. The ability to rate three-dimensional materi-

als, such as Spanish tiles, has been a problem in the past, but that issue is being addressed by the CRRC with funding from PIER.

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The cool-colors program is a collaborative research and development project between PIER, the roofing industry, and the Oak Ridge and Lawrence Berkeley National Laboratories.

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For more information:

For more information about the PIER-funded cool-colors project, visit <http://coolcolors.lbl.gov>.

For a list of Energy Star-labeled roof products, visit www.energystar.gov/index.cfm?c=roof_prods.prroof_products.

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